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Determining the presence of pathogens in food products

CH5200 - Food Microbiology
Experiment 4



Group 2		
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1 General Introduction

Foodborne illness accounts for 600 million illnesses, 420,000 deaths, and the loss of 33 million healthy years of life globally annually (Havelaar et al., 2015). The largest culprits of foodborne disease are foodborne pathogens, making the detection of such pathogens an important field of development. In this experiment, the presence of common foodborne pathogens—*Listeria spp.*, *Salmonella spp.*, and *Campylobacter spp.*—was investigated in common food products, including raw meats and vegetables, to assess the health risks present in a diverse spectrum of food products.

2 Materials and Methods

In this experiment, a 25g sample of frozen chicken fillet was investigated for the presence of *Campylobacter spp.* by using various biochemical tests and an immunological assay (Singlepath®). The chicken used was pre-cut and packaged on 26th April, bought on 27th April from a grocer, and stored in a home freezer until it was thawed for testing on 15th May. Additionally, 25g samples of broccoli from a home refrigerator (bought on 10th May) were tested on the same day as the chicken for the presence of *Salmonella spp.*, *Listeria spp.*, and *L. monocytogenes* using a diverse spectrum of qualitative tests such as selective culturing, isolation, and confirmatory tests on various media.

3 Results and Discussion

This experiment evaluated the presence of pathogenic microorganisms, specifically *Campylobacter spp.*, *Listeria spp.*, *L. monocytogenes*, and *Salmonella spp.*, using a variety of techniques.

Table 1: Presence of pathogens in broccoli and frozen chicken fillet samples

Product	<i>Campylobacter spp.</i>	<i>L. monocytogenes</i>	<i>Listeria spp.</i> other than <i>monocytogenes</i>	<i>Salmonella spp.</i>	<i>Salmonella</i> serology group
Chicken fillet (Frozen)	-	Not done	Not done	Not done	Not done
Broccoli	Not done	+	+	+	Non-specific, Vi

From Table 1, it was concluded that *Campylobacter spp.* was not detected in the frozen chicken fillet sample, whereas *Listeria spp.*, *L. monocytogenes*, and *Salmonella spp.* (non-specific and Vi serology) were detected in the broccoli sample.

3.1. Detection of *Campylobacter spp.* in frozen chicken fillet sample

Enriched cultures of the chicken sample were streaked on selective mCCDA and CASA plates to detect the presence of *Campylobacter spp.* After incubation, the plates obtained showed no strong colony differences (Annex Table A1) between the chicken sample and the positive control (red colonies for CASA and gray colonies for mCCDA). Colonies were then tested to be oxidase and catalase positive, which is typical of *Campylobacter*.

However, the Singlepath® immunological assay revealed a negative result for the presence of *Campylobacter* in the chicken sample. Upon microscopic visualization, the colonies from the chicken sample were observed to be rod-shaped with a distinctively different morphology from the positive control (Annex Tables A3 and A4). Hence, the chicken sample was interpreted to not contain any detectable *Campylobacter*.

3.2. Detection of *L. monocytogenes* in broccoli sample

To detect the presence of *Listeria spp.* and *L. monocytogenes*, an enrichment culture from the broccoli sample was plated on selective ALOA plates.

From Annex Figure B1, both the positive control and broccoli sample culture showed the presence of blue-green colonies with an opaque halo around the colonies. This indicated the presence of both *Listeria spp.* and *L. monocytogenes* in the broccoli sample.

3.3. Detection of *Salmonella* spp. in broccoli sample

From Annex Table C1, broccoli samples enriched in MSRV and MKTTn media showed indicators of *Salmonella* spp.—gray-white motility zones with colonies radiating outward from the center of the droplet on MSRV plates, pale blue/ gray, turbid MKTTn solution with precipitate observed at the bottom of the tube without mixing. Upon culturing with BGA, pink colonies were observed on both MSRV and MKTTn samples. Compared to culturing with XLD, the MKTTn sample showed black colonies (positive), while the MSRV sample showed no black colonies (negative).

Three out of the four samples showed indicators of *Salmonella* spp.—MSRV BGA, MKTTn BGA and MKTTn XLD—which are then further tested to confirm the presence/ absence of *Salmonella* spp.

From Annex Table C3, all three samples used showed positive results for the TSI test, indicating the presence of *Salmonella*—the yellow slant and black butt of the TSI agar indicating glucose fermentation, along with gas formation and H₂S production shown by cracks, bubbles, and blackening of the agar. Similarly, all three samples displayed positive results for the latex agglutination test—red agglutination for MKTTn BGA suggesting Vi serogroup, and grey/ purple agglutination for MSRV BGA and MKTTn XLD suggesting a non-specific *Salmonella* serogroup.

However, for the LDB test, only MSRV BGA and MKTTn XLD were found to be positive—turbid purple media, indicating the presence of *Salmonella*. Additionally, for the UA test, only the MSRV BGA sample resulted in a positive result—unchanged yellow slant, indicating the presence of *Salmonella*. The inconsistent results may be due to differences in colony selection, varying test sensitivity, or metabolic variability and viability across *Salmonella* strains. Hence, the broccoli sample was interpreted to contain *Salmonella* spp.

3.4. Pathogenic diversity across different foods

Table 2 shows the summarized results for the presence of detected pathogens across different food samples (Annex Table D1).

Table 2: Summary of detected pathogens found on tested food products.

Product category	Remarks
Chicken	<i>Campylobacter spp.</i> , <i>Salmonella spp.</i> (C, E or G)
Pork	<i>Salmonella spp.</i> (Non specific, B)
Beef	No pathogens detected
Vegetable (Broccoli)	<i>L. monocytogenes</i> , <i>Salmonella spp.</i> (Non specific, Vi)
Seafood (Blood cockles)	No <i>Listeria spp.</i> detected; not tested for other pathogens.

From the results, *Salmonella spp.* was the most commonly identified pathogen and was detected in raw chicken, pork, and vegetable products. *Salmonella*, a rather common pathogenic bacteria, can infect humans through contaminated food products (e.g., raw meat, vegetables, and dairy products), environmental sources (e.g., water sources), and direct contact with human or animal feces (Huoy et al., 2024). Hence, the presence of *Salmonella* was expected due to the products' freshness and pre-processing conditions. However, *Salmonella* was unexpectedly absent from some raw meat samples (beef, chicken and pork). This may be due to the effect of freezing, which can damage *Salmonella* such that it is unable to sufficiently recover in the 24-hour non-specific enrichment that was done (Digirolamo et al., 1970).

Similarly, *Campylobacter spp.* was unexpectedly absent from some raw chicken samples. This can be attributed to the sensitivity of *Campylobacter* to freezing, chilling or atmospheric air. Such conditions may reduce its viability. It is still possible that *Campylobacter spp.* is present in the samples that tested negative but in a damaged/stressed form, requiring a longer recovery period to be detected (Adams et al., 2024).

L. monocytogenes was only detected in broccoli and absent in all raw and processed meats. *L. monocytogenes* is sensitive to low pH environments; it has a narrow pH range (6.0-7.0) for optimal growth. Hence, the post-rigor mortis drop in pH in meat products may selectively suppress the growth of *L. monocytogenes* compared to other microbes (Adams et al., 2024). As broccoli has a higher pH than raw meats, *L. monocytogenes* flourishes in it.

Blood cockles, as filter-feeding bivalves, can accumulate contaminants, including bacteria, viruses, and toxins. However, the absence of *L. monocytogenes* may be due to its low concentration in water bodies or limited uptake by the cockles. Additionally, depuration (i.e., a post-harvest cleansing process of bivalves) may have removed contaminants, including *L. monocytogenes* (Teo, 2023).

4 Conclusions and Recommendations

Often so, at least one of the pathogens *Salmonella*, *Campylobacter* and *L. monocytogenes* was detected in every food product tested, suggesting that cooking raw foods is still of utmost importance in food safety. *Salmonella* and *Campylobacter* were found to be less ubiquitous than expected, possibly due to freezing. Additionally, *L. monocytogenes* was found to be less prevalent in meats but present in vegetables. Overall, the methods used for isolation and detection were acceptable albeit time-consuming and potentially less sensitive for frozen foods. Hence, it is recommended that longer enrichment times or other methods be used for frozen samples.

The detection of foodborne pathogens still remains as a crucial step in ensuring food safety and security on a national and global scale. This promotes public health and awareness, preventing foodborne outbreaks and ensuring regulatory compliance in the food industry.

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Annex

Table A1: Results of selective enrichment of frozen chicken fillet sample processed using a stomacher (in Modified Charcoal Cefoperazone Desoxycholate Agar [mCCDA] or *Campylobacter* Selective Agar Biomérieux [CASA]); resulting colonies were then isolated on Columbia Agar Base (CAB) plates

Phase	Sample	Culture medium	Observation	Result
Selective enrichment	Positive control	mCCDA	Flat gray colonies present	Positive
		CASA	Red colonies present	Positive
	Chicken sample	mCCDA	Flat gray colonies present	Positive
		CASA	Red colonies present	Positive
Final isolates	mCCDA (Positive control)	CAB	Flat gray colonies present	N/A
	CASA (Positive control)	CAB	Flat gray colonies present	N/A
	mCCDA (Chicken sample)	CAB	Flat gray colonies present	N/A
	CASA (Chicken sample)	CAB	Flat gray colonies present	N/A

Table A2: Images of selective enrichment results from Table A1

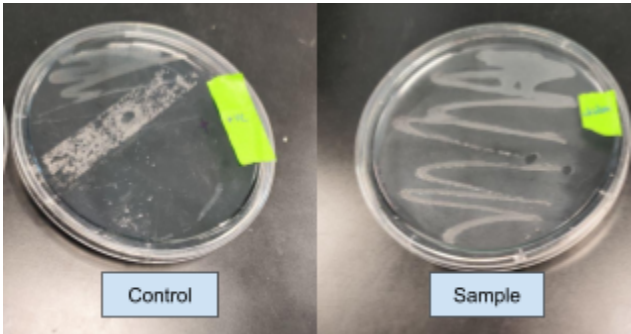
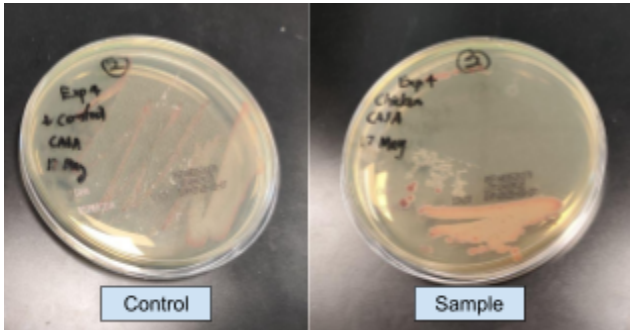
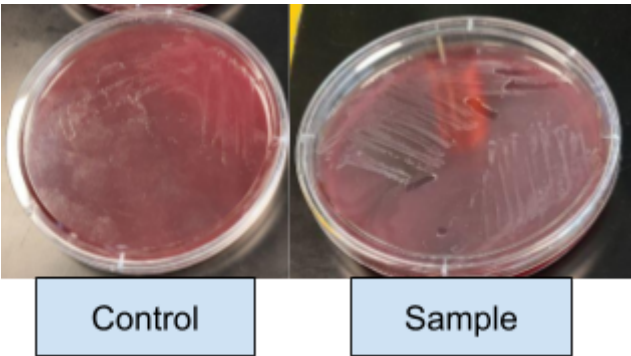
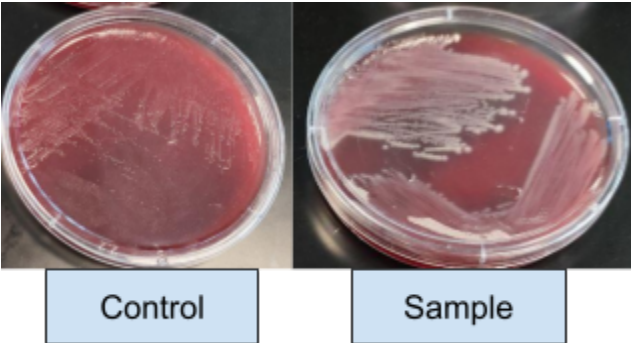
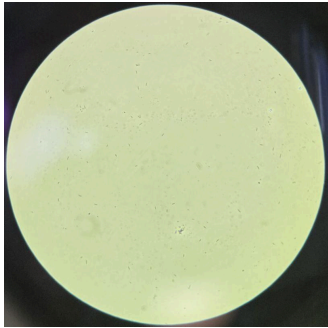
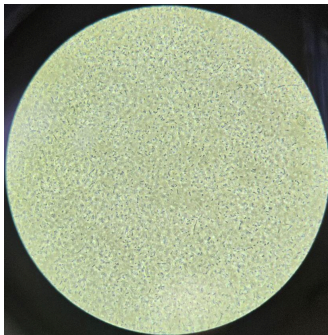
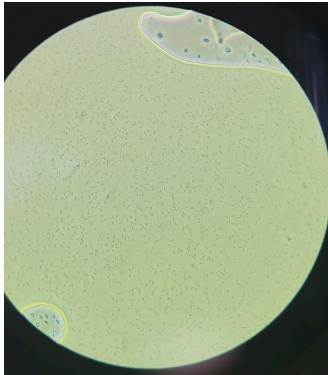
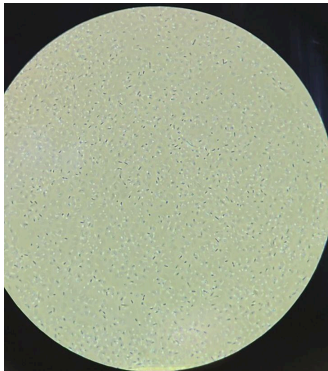
Phase	Culture medium	Results	
Selective enrichment	mCCDA	 Control Sample	
	CASA	 Control Sample	
Final isolates	mCCDA CAB	 Control Sample	
	CASA CAB	 Control Sample	

Table A3: Bacterial identification results of positive controls and chicken samples cultured on mCCDA and CASA media, including oxidase and catalase tests, assessment of cell shape and motility pattern, and Singlepath® testing

Sample/ Test	Oxidase	Catalase	Shape	Motility	Singlepath®	Result
mCCDA (Positive control)	Positive	Positive	Rod	Motile without corkscrew shape	Positive	Positive
CASA (Positive control)	Positive	Positive	Rod	Motile without corkscrew shape	Positive	Positive
mCCDA (Chicken sample)	Positive	Positive	Rod	Motile without corkscrew shape	Negative	Negative
CASA (Chicken sample)	Positive	Positive	Rod	Motile without corkscrew shape	Negative	Negative

Table A4: Images of confirmatory test results from Table A3

Sample/ Test	Microscopy	Singlepath®
mCCDA (Positive control)		
mCCDA (Chicken sample)		[control]: Two lines observed (Positive) [sample]: One line observed (Negative)
CASA (Positive control)		
CASA (Chicken sample)		[control]: Two lines observed (Positive) [sample]: One line observed (Negative)

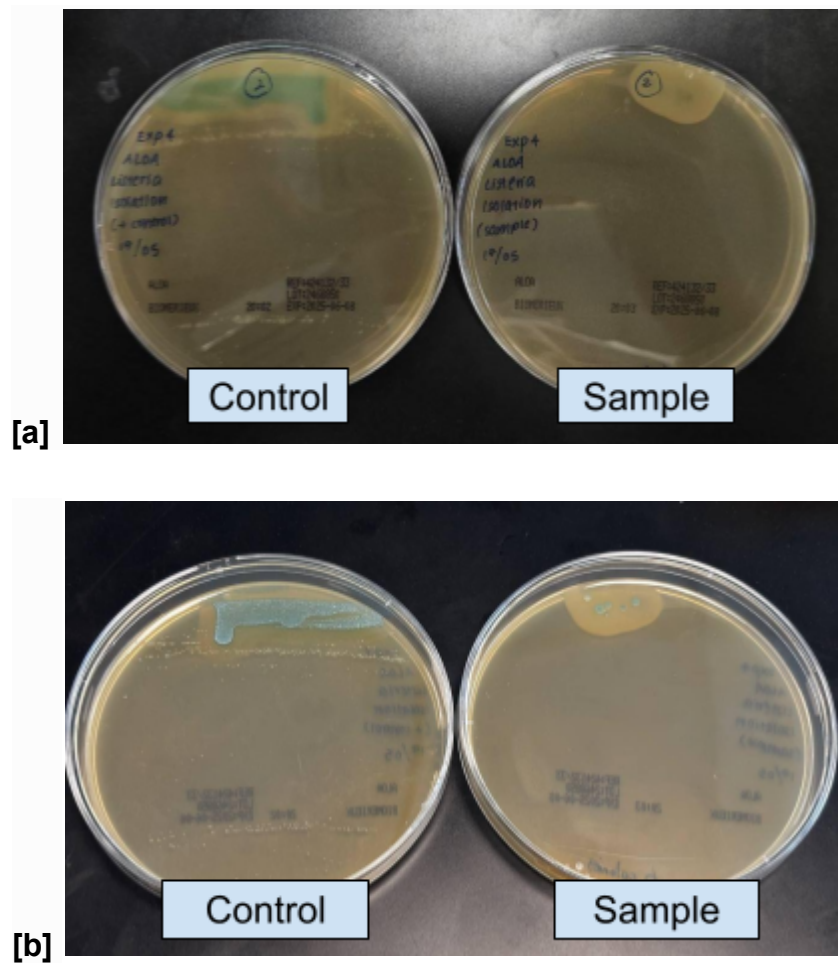


Figure B1: [a] Bottom view of Agar *Listeria* Ottaviani Agosti (ALOA) plates, [right] Top view of ALOA plates (clearer view of blue-green colonies and surrounding halo)

Table C1: Results of selective enrichment of broccoli sample processed using a stomacher (in Modified Semi-solid Rappaport-Vassiliadis [MSRV] or Muller-Kauffmann tetrathionate-novobiocin broth [MKTTn]); resulting colonies were then isolated on culture media (Brilliant Green Agar [BGA] or Xylose Lysine Desoxycholate [XLD])

Sample	Sample used for isolation	Observation	Culture medium	Observation	Result
Positive control	MSRV	Gray-white motility zones, with colonies radiating outward from the center of the droplet	BGA	Pink colonies on cerise medium	Positive
	MKTTn	Pale blue/ gray, turbid solution, with precipitate observed at the bottom of the tube without mixing	BGA	Pink colonies on cerise medium	Positive
	MSRV	Gray-white motility zones, with colonies radiating outward from the center of the droplet	XLD	No black colonies on cerise medium	Positive
	MKTTn	Pale blue/ gray, turbid solution, with precipitate observed at the bottom of the tube without mixing	XLD	Black colonies on cerise medium	Positive
Broccoli sample	MSRV	Gray-white motility zones, with colonies radiating outward from the center of the droplet	BGA	Pink colonies on cerise medium	Positive
	MKTTn	Pale blue/ gray, turbid solution, with precipitate observed at the bottom of the tube without mixing	BGA	Pink colonies on cerise medium	Positive
	MSRV	Gray-white motility zones, with colonies radiating outward from the center of the droplet	XLD	No black colonies on cerise medium	Negative
	MKTTn	Pale blue/ gray, turbid solution, with precipitate observed at the bottom of the tube without mixing	XLD	Black colonies on cerise medium	Positive

Table C2: Images of selective enrichment results from Table C1

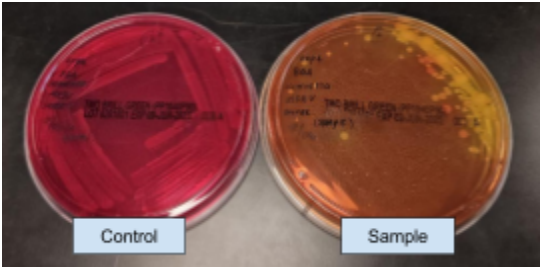
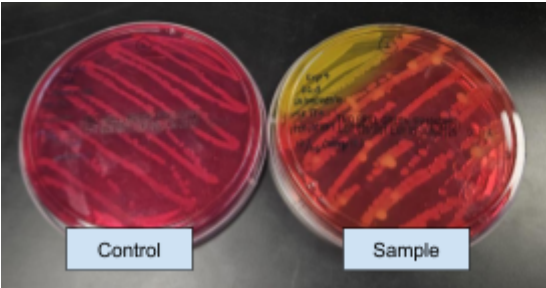
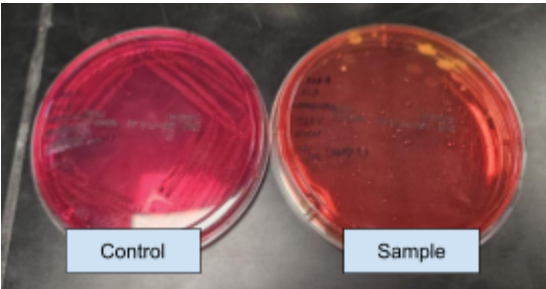
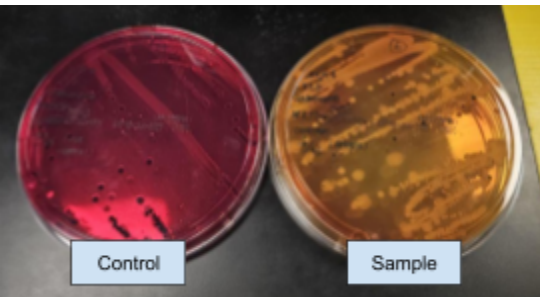
Sample used for isolation	Culture medium	Results
MSRV	BGA	
MKTTn	BGA	
MSRV	XLD	
MKTTn	XLD	

Table C3: Results of the confirmation tests using positive colonies from Table 1—MSRV BGA, MKTTn BGA and MKTTn XLD

Sample/ test		Triple Sugar Iron Agar (TSI)	Lysine Decarboxylase Broth (LDB)	Urea Agar (UA)	Latex agglutination
		Observation	Observation	Observation	Observation
Positive control	MSRV BGA	Red slant and black butt (Positive)	Purple and turbid (Positive)	Yellow slant (Positive)	Not done
	MKTTn BGA	Red slant and black butt (Positive)	Purple and turbid (Positive)	Yellow slant (Positive)	Not done
	MKTTn XLD	Slight yellow slant and black butt (Positive)	Purple and turbid (Positive)	Yellow slant (Positive)	Not done
Broccoli sample	MSRV BGA	Yellow slant and black butt (Positive)	Purple and turbid (Positive)	Yellow slant (Positive)	R1: Grey purple agglutination R2: Grey purple agglutination (Positive)
	MKTTn BGA	Yellow slant and black butt (Positive)	Yellow and turbid (Negative)	Purple slant (Negative)	R1: Grey purple agglutination R2: Red purple agglutination (Positive)
	MKTTn XLD	Slight yellow slant and black butt (Positive)	Purple and turbid (Positive)	Purple slant (Negative)	R1: Grey purple agglutination R2: Grey purple agglutination (Positive)

Table C4: Images of confirmation test results from Table C3

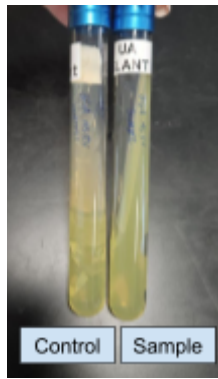
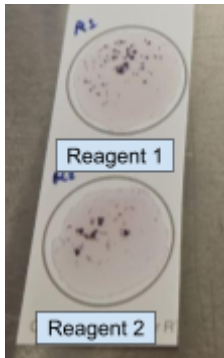
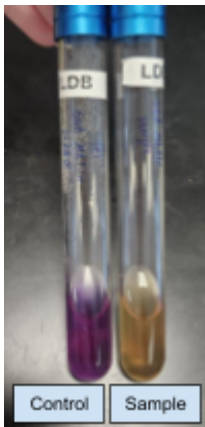
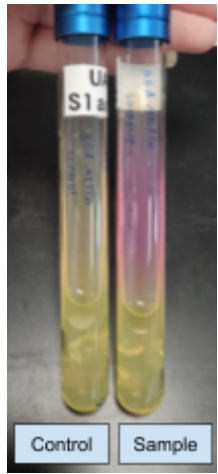
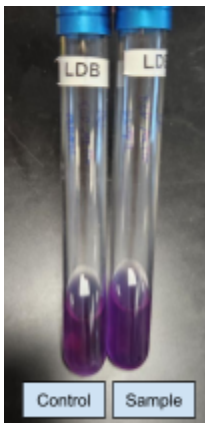
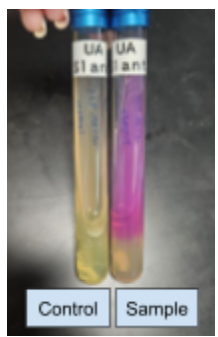
Sample used and culture medium/ Confirmation test results	Triple Sugar Iron Agar (TSI)	Lysine Decarboxylase Broth (LDB)	Urea Agar (UA)	Latex Agglutination
MSRV BGA	 Control Sample	 Control Sample	 Control Sample	 Reagent 1 Reagent 2
MKTTn BGA	 Control Sample	 Control Sample	 Control Sample	 Reagent 1 Reagent 2
MKTTn XLD	 Control Sample	 Control Sample	 Control Sample	 Reagent 1 Reagent 2

Table D1: Presence of the *Campylobacter* spp., *Listeria* spp., and *Salmonella* spp. in common food products.

Category 1: Chicken products					
Product	<i>Campylobacter</i> spp.	<i>L. monocytogenes</i>	<i>Listeria</i> spp. other than monocytogenes	<i>Salmonella</i> spp.	<i>Salmonella</i> serology group
Chicken minced	-	-	+	+	E or G
Chicken liver	+	Not done	Not done	Not done	Not done
Chicken fillet (Frozen)	-	Not done	Not done	Not done	Not done
Chicken thigh	+	-	+	-	-
Chicken heart	Not done	Not done	Not done	-	-
Chicken wing	-	-	+	+	C, E or G
Chicken egg	-	Not done	Not done	Not done	-
Category 2: Pork products					
Product	<i>Campylobacter</i> spp.	<i>L. monocytogenes</i>	<i>Listeria</i> spp. other than monocytogenes	<i>Salmonella</i> spp.	<i>Salmonella</i> serology group
Pork minced 1	-	-	+	+	B
Pork minced 2	Not done	Not done	Not done	-	-
Pork liver	-	-	-	+	Non specific
Pig intestine	-	-	-	-	-
Picnic ham	Not done	-	-	Not done	Not done
Category 3: Beef products					
Product	<i>Campylobacter</i> spp.	<i>L. monocytogenes</i>	<i>Listeria</i> spp. other than monocytogenes	<i>Salmonella</i> spp.	<i>Salmonella</i> serology group
Beef minced	-	-	-	-	-

Category 4: Others					
Product	<i>Campylobacter</i> <i>spp.</i>	<i>L.</i> <i>monocytogenes</i>	<i>Listeria</i> <i>spp.</i> other than monocytogenes	<i>Salmonella</i> <i>spp.</i>	<i>Salmonella</i> serology group
Broccoli	Not done	+	+	+	Vi, Non specific
Blood cockles	Not done	-	-	Not done	Not done